

Number Theory and Cryptography

Chapter 4



Chapter Motivation

- *Number theory* is the part of mathematics devoted to the study of the integers and their properties.
- Key ideas in number theory include divisibility and the primality of integers.
- Representations of integers, including binary and hexadecimal representations, are part of number theory.
- Number theory has long been studied because of the beauty of its ideas, its accessibility, and its wealth of open questions.
- We'll use many ideas developed in Chapter 1 about proof methods and proof strategy in our exploration of number theory.
- Mathematicians have long considered number theory to be pure mathematics, but it has important applications to computer science and cryptography studied in Sections 4.5 and 4.6.



Chapter Summary

- Divisibility and Modular Arithmetic
- Integer Representations and Algorithms
- Primes and Greatest Common Divisors (公约数)
- Solving Congruences (同余方程)
- Applications of Congruences
- Cryptography

Divisibility and Modular Arithmetic

Section 4.1



Section Summary

- Division
- Division Algorithm
- Modular Arithmetic

Division

Definition: If a and b are integers with $a \neq 0$, then a divides b if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor* or *divisor* of b and that b is a multiple of a .
- The notation $a \mid b$ denotes that a divides b .
- If $a \mid b$, then b/a is an integer.
- If a does not divide b , we write $a \nmid b$.

Example: Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Properties of Divisibility

Theorem 1: Let a , b , and c be integers, where $a \neq 0$.

- i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- ii. If $a \mid b$, then $a \mid bc$ for all integers c ;
- iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t). \quad \text{Hence, } a \mid (b + c)$$



(Exercises 3 and 4 ask for proofs of parts (ii) and (iii).)

Corollary: If a , b , and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Can you show how it follows easily from (ii) and (i) of Theorem 1?

Division Algorithm

- When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the “Division Algorithm,” but is really a theorem.

Division Algorithm: If a is an integer and d a positive integer, then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$ (proved in Section 5.2).

- d is called the *divisor*.
- a is called the *dividend*.
- q is called the *quotient*.
- r is called the *remainder*.

Definitions of Functions
div and **mod**

$$q = a \text{ div } d$$
$$r = a \text{ mod } d$$

Examples:

- What are the quotient and remainder when 101 is divided by 11?

Solution: The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.

- What are the quotient and remainder when -11 is divided by 3?

Solution: The quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Congruence Relation

Definition: If a and b are integers and m is a positive integer, then a is *congruent* to b modulo m if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus*.
- Two integers are congruent mod m if and only if they have the same remainder when divided by m .
- If a is not congruent to b modulo m , we write

$$a \not\equiv b \pmod{m}$$

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since 6 divides $24 - 14 = 10$ is not divisible by 6.

More on Congruences

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$. Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$.



The Relationship between $(\text{mod } m)$ and $\text{mod } m$ Notations

- The use of “mod” in $a \equiv b \pmod{m}$ and $a \text{ mod } m = b$ are different.
 - $a \equiv b \pmod{m}$ is a relation on the set of integers.
 - In $a \text{ mod } m = b$, the notation **mod** denotes a function.
- The relationship between these notations is made clear in this theorem.
- **Theorem 3:** Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \text{ mod } m = b \text{ mod } m$. (*Proof in the exercises*)

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. ◀

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 * 11 \equiv 2 * 1 = 2 \pmod{5}$$

Algebraic Manipulation of Congruences

- Multiplying both sides of a valid congruence by an integer preserves validity.
If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.
- Adding an integer to both sides of a valid congruence preserves validity.
If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.
- Dividing a congruence by an integer does not always produce a valid congruence.

Example: The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

See Section 4.3 for conditions when division is ok.

Computing the **mod** m Function of Products and Sums

- We use the following corollary to Theorem 5 to compute the remainder of the product or sum of two integers when divided by m from the remainders when each is divided by m .

Corollary: Let m be a positive integer and let a and b be integers. Then

$$(a + b) \text{ (mod } m) = ((a \text{ mod } m) + (b \text{ mod } m)) \text{ mod } m$$

and

$$ab \text{ mod } m = ((a \text{ mod } m) (b \text{ mod } m)) \text{ mod } m.$$

(proof in text)

Arithmetic Modulo m

Definitions: Let Z_m be the set of nonnegative integers less than m : $\{0, 1, \dots, m-1\}$

- The operation $+_m$ is defined as $a +_m b = (a + b) \bmod m$. This is *addition modulo m* .
- The operation $\cdot_m$ is defined as $a \cdot_m b = (a \cdot b) \bmod m$. This is *multiplication modulo m* .
- Using these operations is said to be doing *arithmetic modulo m* .

Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.

Solution: Using the definitions above:

- $7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$
- $7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$

Arithmetic Modulo m

- The operations $+_m$ and $\cdot_m$ satisfy many of the same properties as ordinary addition and multiplication.
 - *Closure*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b$ and $a \cdot_m b$ belong to $\mathbf{Z}_m$.
 - *Associativity*: If a , b , and c belong to $\mathbf{Z}_m$, then $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
 - *Commutativity*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$.
 - *Identity elements*: The elements 0 and 1 are identity elements for addition and multiplication modulo m , respectively.
 - If a belongs to $\mathbf{Z}_m$, then $a +_m 0 = a$ and $a \cdot_m 1 = a$.

continued $\rightarrow$

Arithmetic Modulo m

- *Additive inverses:* If $a \neq 0$ belongs to $\mathbf{Z}_m$, then $m - a$ is the additive inverse of a modulo m and 0 is its own additive inverse.
 - $a +_m (m - a) = 0$ and $0 +_m 0 = 0$
- *Distributivity:* If a , b , and c belong to $\mathbf{Z}_m$, then
 - $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$ and $(a +_m b) \cdot_m c = (a \cdot_m c) +_m (b \cdot_m c)$.
- Exercises 42-44 ask for proofs of these properties.
- Multiplicative inverses have not been included since they do not always exist. For example, there is no multiplicative inverse of 2 modulo 6.
- (*optional*) Using the terminology of abstract algebra, $\mathbf{Z}_m$ with $+_m$ is a commutative group and $\mathbf{Z}_m$ with $+_m$ and $\cdot_m$ is a commutative ring.



Exercise

- Sec4.1 47

Integer Representations and Algorithms

Section 4.2



Section Summary

- Integer Representations
 - Base b Expansions
 - Binary Expansions
 - Octal Expansions
 - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations

Representations of Integers

- In the modern world, we use *decimal*, or *base 10, notation* to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b , where b is a positive integer greater than 1.
- The bases $b = 2$ (*binary*), $b = 8$ (*octal*), and $b = 16$ (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

Base b Representations

- We can use positive integer b greater than 1 as a base, because of this theorem:

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$. The $a_j, j = 0, \dots, k$ are called the base- b digits of the representation.

(We will prove this using mathematical induction in Section 5.1.)

- The representation of n given in Theorem 1 is called the *base b expansion of n* and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.
- We usually omit the subscript 10 for base 10 expansions.

Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

Solution:

$$(1\ 0101\ 1111)_2 = 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ = 351.$$

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

Octal Expansions

The octal expansion (base 8) uses the digits $\{0,1,2,3,4,5,6,7\}$.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example: What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

Hexadecimal Expansions

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits $\{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F\}$. The letters A through F represent the decimal numbers 10 through 15.

Example: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution:

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

Example: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution: $1 \cdot 16^2 + 14 \cdot 16^1 + 5 \cdot 16^0 = 256 + 224 + 5 = 485$

Base Conversion

To construct the base b expansion of an integer n :

- Divide n by b to obtain a quotient and remainder.

$$n = bq_0 + a_0 \quad 0 \leq a_0 \leq b$$

- The remainder, a_0 , is the rightmost digit in the base b expansion of n . Next, divide q_0 by b .

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 \leq b$$

- The remainder, a_1 , is the second digit from the right in the base b expansion of n .
- Continue by successively dividing the quotients by b , obtaining the additional base b digits as the remainder. The process terminates when the quotient is 0.

continued →

Algorithm: Constructing Base b Expansions

```
procedure base b expansion( $n, b$ : positive integers with  $b > 1$ )  
   $q := n$   
   $k := 0$   
  while ( $q \neq 0$ )  
     $a_k := q \bmod b$   
     $q := q \div b$   
     $k := k + 1$   
  return( $a_{k-1}, \dots, a_1, a_0$ ) {( $a_{k-1} \dots a_1 a_0$ ) $b$  is base  $b$  expansion of  $n$ }
```

- q represents the quotient obtained by successive divisions by b , starting with $q = n$.
- The digits in the base b expansion are the remainders of the division given by $q \bmod b$.
- The algorithm terminates when $q = 0$ is reached.

Base Conversion

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

- $12345 = 8 \cdot 1543 + 1$
- $1543 = 8 \cdot 192 + 7$
- $192 = 8 \cdot 24 + 0$
- $24 = 8 \cdot 3 + 0$
- $3 = 8 \cdot 0 + 3$

The remainders are the digits from right to left yielding $(30071)_8$.

Comparison of Hexadecimal, Octal, and Binary Representations

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Initial 0s are not shown

Each octal digit corresponds to a block of 3 binary digits.

Each hexadecimal digit corresponds to a block of 4 binary digits.

So, conversion between binary, octal, and hexadecimal is easy.

Conversion Between Binary, Octal, and Hexadecimal Expansions

Example: Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.

Solution:

- To convert to octal, we group the digits into blocks of three $(011\ 111\ 010\ 111\ 100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, 7, 2, 7, and 4. Hence, the solution is $(37274)_8$.
- To convert to hexadecimal, we group the digits into blocks of four $(0011\ 1110\ 1011\ 1100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, E, B, and C. Hence, the solution is $(3EBC)_{16}$.

Binary Addition of Integers

- Algorithms for performing operations with integers using their binary expansions are important as computer chips work with binary numbers. Each digit is called a *bit*.

```
procedure add( $a, b$ : positive integers)
{the binary expansions of  $a$  and  $b$  are  $(a_{n-1}, a_{n-2}, \dots, a_0)_2$  and  $(b_{n-1}, b_{n-2}, \dots, b_0)_2$ , respectively}
 $c := 0$ 
for  $j := 0$  to  $n - 1$ 
     $d := \lfloor (a_j + b_j + c) / 2 \rfloor$ 
     $s_j := a_j + b_j + c - 2d$ 
     $c := d$ 
 $s_n := c$ 
return  $(s_0, s_1, \dots, s_n)$  {the binary expansion of the sum is  $(s_n, s_{n-1}, \dots, s_0)_2$ }
```

- The number of additions of bits used by the algorithm to add two n -bit integers is $O(n)$.

Binary Multiplication of Integers

- Algorithm for computing the product of two n bit integers.

```
procedure multiply( $a, b$ : positive integers)
{the binary expansions of  $a$  and  $b$  are  $(a_{n-1}, a_{n-2}, \dots, a_0)_2$  and  $(b_{n-1}, b_{n-2}, \dots, b_0)_2$ , respectively}
for  $j := 0$  to  $n - 1$ 
    if  $b_j = 1$  then  $c_j = a$  shifted  $j$  places
    else  $c_j := 0$ 
{ $c_0, c_1, \dots, c_{n-1}$  are the partial products}
 $p := 0$ 
for  $j := 0$  to  $n - 1$ 
     $p := p + c_j$ 
return  $p$  { $p$  is the value of  $ab$ }
```

- The number of additions of bits used by the algorithm to multiply two n -bit integers is $O(n^2)$.

Binary Modular Exponentiation

- In cryptography, it is important to be able to find $b^n \bmod m$ efficiently, where b , n , and m are large integers.
- Use the binary expansion of n , $n = (a_{k-1}, \dots, a_1, a_0)_2$, to compute b^n .

Note that:

$$b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2 + a_0} = b^{a_{k-1} \cdot 2^{k-1}} \dots b^{a_1 \cdot 2} \cdot b^{a_0}.$$

Therefore, to compute b^n , we need only compute the values of b , b^2 , $(b^2)^2 = b^4$, $(b^4)^2 = b^8$, ..., b^{2^k} and then multiply the terms b^{2^j} in this list, where $a_j = 1$.

Example: Compute 3^{11} using this method.

Solution: Note that $11 = (1011)_2$ so that $3^{11} = 3^8 3^2 3^1 = ((3^2)^2)^2 3^2 3^1 = (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 117,147$.

continued →

Binary Modular Exponentiation Algorithm

- The algorithm successively finds $b \bmod m$, $b^2 \bmod m$, $b^4 \bmod m$, ..., $b^{2^{k-1}} \bmod m$, and multiplies together the terms b^{2^j} where $a_j = 1$.

```
procedure modular_exponentiation( $b$ : integer,  $n = (a_{k-1}a_{k-2}\dots a_1a_0)_2$ ,  $m$ : positive integers)
 $x := 1$ 
 $power := b \bmod m$ 
for  $i := 0$  to  $k - 1$ 
    if  $a_i = 1$  then  $x := (x \cdot power) \bmod m$ 
     $power := (power \cdot power) \bmod m$ 
return  $x$  { $x$  equals  $b^n \bmod m$ }
```

- $O((\log m)^2 \log n)$ bit operations are used to find $b^n \bmod m$.

Binary Modular Exponentiation

Example: Compute $3^{644} \bmod 645$ using this method.

Solution:

$i=0$: 因 $a_0=0$, 有 $x=1$ 和 $power=3^2 \bmod 645=9 \bmod 645=9$;
 $i=1$: 因 $a_1=0$, 有 $x=1$ 和 $power=9^2 \bmod 645=81 \bmod 645=81$;
 $i=2$: 因 $a_2=1$, 有 $x=1 \cdot 81 \bmod 645=81$ 和 $power=81^2 \bmod 645=6561 \bmod 645=111$;
 $i=3$: 因 $a_3=0$, 有 $x=81$ 和 $power=111^2 \bmod 645=12\,321 \bmod 645=66$;
 $i=4$: 因 $a_4=0$, 有 $x=81$ 和 $power=66^2 \bmod 645=4356 \bmod 645=486$;
 $i=5$: 因 $a_5=0$, 有 $x=81$ 和 $power=486^2 \bmod 645=236\,196 \bmod 645=126$;
 $i=6$: 因 $a_6=0$, 有 $x=81$ 和 $power=126^2 \bmod 645=15\,876 \bmod 645=396$;
 $i=7$: 因 $a_7=1$, 有 $x=(81 \cdot 396) \bmod 645=471$ 和 $power=396^2 \bmod 645=156\,816 \bmod 645=81$;
 $i=8$: 因 $a_8=0$, 有 $x=471$ 和 $power=81^2 \bmod 645=6561 \bmod 645=111$;
 $i=9$: 因 $a_9=1$, 有 $x=(471 \cdot 111) \bmod 645=36$ 。



Exercise

- Sec 4.2 25,31

Primes and Greatest Common Divisors

Section 4.3



Section Summary

- Prime Numbers and their Properties
- Conjectures and Open Problems About Primes
- Greatest Common Divisors and Least Common Multiples
- The Euclidian Algorithm
- gcds as Linear Combinations



Primes

Definition: A positive integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*.

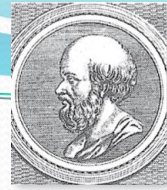
Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

The Fundamental Theorem of Arithmetic

Theorem: Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

Examples:

- [illegible]



Eratosthenes
(276-194 B.C.)

The Sieve of Eratosthenes

- The *Sieve of Eratosthenes* can be used to find all primes not exceeding a specified positive integer. For example, begin with the list of integers between 1 and 100.
 - a. Delete all the integers, other than 2, divisible by 2.
 - b. Delete all the integers, other than 3, divisible by 3.
 - c. Next, delete all the integers, other than 5, divisible by 5.
 - d. Next, delete all the integers, other than 7, divisible by 7.
 - e. Since all the remaining integers are not divisible by any of the previous integers, other than 1, the primes are:
 $\{2, 3, 7, 11, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97\}$

continued →

The Sieve of Eratosthenes

TABLE 1 The Sieve of Eratosthenes.

Integers divisible by 2 other than 2 receive an underline.										Integers divisible by 3 other than 3 receive an underline.									
1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	9	<u>10</u>	1	2	3	<u>4</u>	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	15	<u>16</u>	17	<u>18</u>	19	<u>20</u>	11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
21	<u>22</u>	23	<u>24</u>	25	<u>26</u>	27	<u>28</u>	29	<u>30</u>	<u>21</u>	<u>22</u>	23	<u>24</u>	25	<u>26</u>	<u>27</u>	<u>28</u>	29	<u>30</u>
31	<u>32</u>	33	<u>34</u>	35	<u>36</u>	37	<u>38</u>	39	<u>40</u>	31	<u>32</u>	<u>33</u>	<u>34</u>	35	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	45	<u>46</u>	47	<u>48</u>	49	<u>50</u>	41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>
51	<u>52</u>	53	<u>54</u>	55	<u>56</u>	57	<u>58</u>	59	<u>60</u>	51	<u>52</u>	53	<u>54</u>	55	<u>56</u>	<u>57</u>	<u>58</u>	59	<u>60</u>
61	<u>62</u>	63	<u>64</u>	65	<u>66</u>	67	<u>68</u>	69	<u>70</u>	61	<u>62</u>	<u>63</u>	<u>64</u>	65	<u>66</u>	67	<u>68</u>	69	<u>70</u>
71	<u>72</u>	73	<u>74</u>	75	<u>76</u>	77	<u>78</u>	79	<u>80</u>	71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
81	<u>82</u>	83	<u>84</u>	85	<u>86</u>	87	<u>88</u>	89	<u>90</u>	<u>81</u>	<u>82</u>	83	<u>84</u>	85	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	93	<u>94</u>	95	<u>96</u>	97	<u>98</u>	99	<u>100</u>	91	<u>92</u>	<u>93</u>	<u>94</u>	95	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>
Integers divisible by 5 other than 5 receive an underline.										Integers divisible by 7 other than 7 receive an underline; integers in color are prime.									
1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	<u>9</u>	<u>10</u>	1	2	3	4	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>	11	<u>12</u>	<u>13</u>	<u>14</u>	<u>15</u>	<u>16</u>	<u>17</u>	<u>18</u>	<u>19</u>	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	<u>25</u>	<u>26</u>	<u>27</u>	<u>28</u>	29	<u>30</u>	<u>21</u>	<u>22</u>	<u>23</u>	<u>24</u>	<u>25</u>	<u>26</u>	<u>27</u>	<u>28</u>	<u>29</u>	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>	<u>31</u>	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	<u>37</u>	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>	<u>41</u>	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	<u>47</u>	<u>48</u>	49	<u>50</u>
<u>51</u>	<u>52</u>	53	<u>54</u>	<u>55</u>	<u>56</u>	57	<u>58</u>	59	<u>60</u>	51	<u>52</u>	<u>53</u>	<u>54</u>	<u>55</u>	<u>56</u>	57	58	<u>59</u>	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	<u>65</u>	<u>66</u>	67	<u>68</u>	69	<u>70</u>	<u>61</u>	<u>62</u>	<u>63</u>	<u>64</u>	<u>65</u>	<u>66</u>	<u>67</u>	68	69	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>	<u>71</u>	<u>72</u>	<u>73</u>	<u>74</u>	<u>75</u>	<u>76</u>	<u>77</u>	<u>78</u>	<u>79</u>	<u>80</u>
<u>81</u>	<u>82</u>	83	<u>84</u>	<u>85</u>	<u>86</u>	87	<u>88</u>	89	<u>90</u>	<u>81</u>	<u>82</u>	<u>83</u>	<u>84</u>	<u>85</u>	<u>86</u>	87	<u>88</u>	<u>89</u>	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	<u>95</u>	<u>96</u>	97	<u>98</u>	99	<u>100</u>	91	<u>92</u>	<u>93</u>	<u>94</u>	<u>95</u>	<u>96</u>	<u>97</u>	<u>98</u>	99	<u>100</u>

If an integer n is a composite integer, then it has a prime divisor less than or equal to $\sqrt{n}$.

To see this, note that if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Trial division, a very inefficient method of determining if a number n is prime, is to try every integer $i \leq \sqrt{n}$ and see if n is divisible by i .

Infinitude of Primes



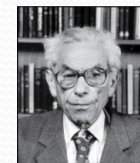
Euclid
(325 B.C.E. – 265 B.C.E.)

Theorem: There are infinitely many primes. (Euclid)

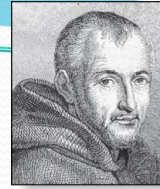
Proof: Assume finitely many primes: $p_1, p_2, \dots, p_n$

- Let $q = p_1 p_2 \cdots p_n + 1$
- Either q is prime or by the fundamental theorem of arithmetic it is a product of primes.
 - But none of the primes p_i divides q since if $p_i \mid q$, then p_i divides $q - p_1 p_2 \cdots p_n = 1$.
 - Hence, there is a prime not on the list $p_1, p_2, \dots, p_n$. It is either q , or if q is composite, it is a prime factor of q . This contradicts the assumption that $p_1, p_2, \dots, p_n$ are all the primes.
- Consequently, there are infinitely many primes.

This proof was given by Euclid *The Elements*. The proof is considered to be one of the most beautiful in all mathematics. It is the first proof in *The Book*, inspired by the famous mathematician Paul Erdős' imagined collection of perfect proofs maintained by God.



Paul Erdős
(1913-1996)



Marin Mersenne
(1588-1648)

Mersenne Primes

Definition: Prime numbers of the form $2^p - 1$, where p is prime, are called *Mersenne primes*.

- $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$, and $2^7 - 1 = 127$ are Mersenne primes.
- $2^{11} - 1 = 2047$ is not a Mersenne prime since $2047 = 23 \cdot 89$.
- There is an efficient test for determining if $2^p - 1$ is prime.
- The largest known prime numbers are Mersenne primes.
- As of mid 2011, 47 Mersenne primes were known, the largest is $2^{43,112,609} - 1$, which has nearly 13 million decimal digits.
- The *Great Internet Mersenne Prime Search (GIMPS)* is a distributed computing project to search for new Mersenne Primes.

<http://www.mersenne.org/>

Distribution of Primes

- Mathematicians have been interested in the distribution of prime numbers among the positive integers. In the nineteenth century, the *prime number theorem* was proved which gives an asymptotic estimate for the number of primes not exceeding x .

Prime Number Theorem: The ratio of the number of primes not exceeding x and $x/\ln x$ approaches 1 as x grows without bound. ($\ln x$ is the natural logarithm of x)

- The theorem tells us that the number of primes not exceeding x , can be approximated by $x/\ln x$.
- The odds that a randomly selected positive integer less than n is prime are approximately $(n/\ln n)/n = 1/\ln n$.

Primes and Arithmetic Progressions (*optional*)

- Euclid's proof that there are infinitely many primes can be easily ^{L1}_{L2} adapted to show that there are infinitely many primes in the following $4k + 3$, $k = 1, 2, \dots$ (See Exercise 55)
- In the 19th century G. Lejuenne Dirchlet showed that every arithmetic progression $ka + b$, $k = 1, 2, \dots$, where a and b have no common factor greater than 1 contains infinitely many primes. (The proof is beyond the scope of the text.)
- Are there long arithmetic progressions made up entirely of primes?
 - 5, 11, 17, 23, 29 is an arithmetic progression of five primes.
 - 199, 409, 619, 829, 1039, 1249, 1459, 1669, 1879, 2089 is an arithmetic progression of ten primes.
- In the 1930s, Paul Erdős conjectured that for every positive integer n greater than 1, there is an arithmetic progression of length n made up entirely of primes. This was proven in 2006, by Ben Green and Terrence Tao.



Terence Tao
(Born 1975 陶哲轩)

- L1** 陶哲轩在7岁进入高中就读，9岁进入大学，10岁、11岁、12岁参加国际数学奥林匹克竞赛，分获铜牌、银牌、金牌。他还未13岁时已赢得国际数学奥林匹克竞赛金牌，这项纪录至今也是由他保持。他在16岁获得学士学位，17岁获得硕士学位，21岁获得普林斯顿大学博士学位。他从24岁起在加利福尼亚大学洛杉矶分校担任教授，成为加利福尼亚大学洛杉矶分校有史以来最年轻的正教授。2006年，31岁时获得数学界的诺贝尔奖“菲尔兹”奖。
Lenovo, 2018/4/10
- L2** 2014年6月荣获被喻为“豪华版诺贝尔奖”的“科学突破奖”的数学奖，奖金高达300万美元。
Lenovo, 2018/4/10

陶哲轩,天才的足迹

- 陶哲轩在7岁进入高中就读，9岁进入大学，10岁、11岁、12岁参加国际数学奥林匹克竞赛，分获铜牌、银牌、金牌。他还未13岁时已赢得国际数学奥林匹克竞赛金牌，这项纪录至今也是由他保持。他在16岁获得学士学位，17岁获得硕士学位，21岁获得普林斯顿大学博士学位。他从24岁起在加利福尼亚大学洛杉矶分校担任教授，成为加利福尼亚大学洛杉矶分校有史以来最年轻的正教授。2006年，31岁时获得数学界的诺贝尔奖“菲尔兹”奖。
- 2014年6月荣获被喻为“豪华版诺贝尔奖”的“科学突破奖”的数学奖，奖金高达300万美元。

Generating Primes

- The problem of generating large primes is of both theoretical and practical interest.
- We will see (in Section 4.6) that finding large primes with hundreds of digits is important in cryptography.
- So far, no useful closed formula that always produces primes has been found. There is no simple function $f(n)$ such that $f(n)$ is prime for all positive integers n .
- But $f(n) = n^2 - n + 41$ is prime for all integers $1, 2, \dots, 40$. Because of this, we might conjecture that $f(n)$ is prime for all positive integers n . But $f(41) = 41^2$ is not prime.
- More generally, there is no polynomial with integer coefficients such that $f(n)$ is prime for all positive integers n . (See supplementary Exercise 23.)
- Fortunately, we can generate large integers which are almost certainly primes. See Chapter 7.

Conjectures about Primes

- Even though primes have been studied extensively for centuries, many conjectures about them are unresolved, including:
- *Goldbach's Conjecture*: Every even integer n , $n > 2$, is the sum of two primes. It has been verified by computer for all positive even integers up to $1.6 \cdot 10^{18}$. The conjecture is believed to be true by most mathematicians.
- Among these are the result that every even integer greater than 2 is the sum of at most six primes (proved in 1995 by O. Ramaré) and that every sufficiently large positive integer is the sum of a prime and a number that is either prime or the product of two primes (proved in 1966 by J. R. Chen). Perhaps Goldbach's conjecture will be settled in the not too distant future.

Conjectures about Primes

- There are infinitely many primes of the form $n^2 + 1$, where n is a positive integer. But it has been shown that there are infinitely many primes of the form $n^2 + 1$, or the product of at most two primes, where n is a positive integer.
- *The Twin Prime Conjecture:* The twin prime conjecture is that there are infinitely many pairs of twin primes. Twin primes are pairs of primes that differ by 2. Examples are 3 and 5, 5 and 7, 11 and 13, 4967 and 4969, etc. The current world's record for twin primes, as of early 2018, consists of the numbers, $2,996,863,034,895 \cdot 2^{1,290,000} \pm 1$, which have 388,342 decimal digits.

Conjectures about Primes

- there are infinitely many twin primes. The strongest result proved concerning twin primes is that there are infinitely many pairs p and $p + 2$, where p is prime and $p + 2$ is prime or the product of two primes (proved by J. R. Chen in 1966)
- Let $P(n)$ be the statement that there are infinitely many pairs of primes that differ by exactly n .
- Yitang Zhang, a 50-year-old professor at the University of New Hampshire, who had not published a paper since 2001, proved the this (named bounded gap) conjecture in 2013. In particular, he showed that there is an integer $N < 70,000,000$ such that $P(N)$ is true.

Yitang Zhang

- was born in Shanghai, China, in 1955. receiving his bachelor's and master's degree in 1982 and 1984, respectively in Peking University. He moved to the United States, attending Purdue University and completing the work for his Ph.D. in 1991.
- 张益唐获得博士学位后，因就业市场不景气及与导师学术意见不合，未能找到学术职位。他曾在纽约皇后区的一家餐馆送餐，后转至肯塔基州朋友的赛百味餐厅工作。他甚至一度在求职期间栖身于自己的车内，但最终成功入职新罕布什尔大学担任讲师。自1999年至2014年初，他一直在该校任教。2009至2013年间，他全心投入到“有界间距猜想”的研究中，每周七天、日均投入约十小时，最终取得关键性突破。这一成就促使新罕布什尔大学将他擢升为教授。2015年，他接受了加州大学圣塔芭芭拉分校的正教授职位。2014年，张益唐荣获麦克阿瑟奖，即俗称的“天才奖”。

Greatest Common Divisor

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

One can find greatest common divisors of small numbers by inspection.

Example: What is the greatest common divisor of 24 and 36?

Solution: $\gcd(24, 36) = 12$

Example: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17, 22) = 1$

Greatest Common Divisor

Definition: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, 10, 17, and 21 are pairwise relatively prime.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, 10, 19, and 24 are not pairwise relatively prime.

Greatest Common Divisor

Definition: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, 10, 17, and 21 are pairwise relatively prime.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, 10, 19, and 24 are not pairwise relatively prime.

Finding the Greatest Common Divisor Using Prime Factorizations

- Suppose the prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where each exponent is a nonnegative integer, and where all primes occurring in either prime factorization are included in both. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}.$$

- This formula is valid since the integer on the right (of the equals sign) divides both a and b . No larger integer can divide both a and b .

Example: $120 = 2^3 \cdot 3 \cdot 5$ $500 = 2^2 \cdot 5^3$

$$\gcd(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$$

- Finding the gcd of two positive integers using their prime factorizations is not efficient because there is no efficient algorithm for finding the prime factorization of a positive integer.

Least Common Multiple

Definition: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . It is denoted by $\text{lcm}(a,b)$.

- The least common multiple can also be computed from the prime factorizations.

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

This number is divided by both a and b and no smaller number is divided by a and b .

Example: $\text{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3,4)} 3^{\max(5,3)} 7^{\max(2,0)} = 2^4 3^5 7^2$

- The greatest common divisor and the least common multiple of two integers are related by:

Theorem 5: Let a and b be positive integers. Then

$$ab = \text{gcd}(a,b) \cdot \text{lcm}(a,b)$$

(*proof is Exercise 31*)



Euclid
(325 B.C.E. – 265 B.C.E.)

Euclidean Algorithm

- The Euclidian algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(a,c)$ when $a > b$ and c is the remainder when a is divided by b .

Example: Find $\gcd(91, 287)$:

- $287 = 91 \cdot 3 + 14$ Divide 287 by 91
- $91 = 14 \cdot 6 + 7$ Divide 91 by 14
- $14 = 7 \cdot 2 + 0$ Divide 14 by 7
Stopping condition

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

continued →

Euclidean Algorithm

- The Euclidean algorithm expressed in pseudocode is:

```
procedure gcd(a, b: positive integers)
```

```
  x := a
```

```
  x := b
```

```
  while y ≠ 0
```

```
    r := x mod y
```

```
    x := y
```

```
    y := r
```

```
  return x {gcd(a,b) is x}
```

- In Section 5.3, we'll see that the time complexity of the algorithm is $O(\log b)$, where $a > b$.

Correctness of Euclidean Algorithm

Lemma 1: Let $a = bq + r$, where a , b , q , and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

Proof:

- Suppose that d divides both a and b . Then d also divides $a - bq = r$ (by Theorem 1 of Section 4.1). Hence, any common divisor of a and b must also be any common divisor of b and r .
- Suppose that d divides both b and r . Then d also divides $bq + r = a$. Hence, any common divisor of a and b must also be a common divisor of b and r .
- Therefore, $\gcd(a, b) = \gcd(b, r)$.



Correctness of Euclidean Algorithm

- Suppose that a and b are positive integers with $a \geq b$.
Let $r_0 = a$ and $r_1 = b$.
Successive applications of the division algorithm yields:

$$\begin{aligned}r_0 &= r_1 q_1 + r_2 & 0 \leq r_2 < r_1, \\r_1 &= r_2 q_2 + r_3 & 0 \leq r_3 < r_2, \\&\vdots \\&\vdots \\&\vdots \\r_{n-2} &= r_{n-1} q_{n-1} + r_n & 0 \leq r_n < r_{n-1}, \\r_{n-1} &= r_n q_n.\end{aligned}$$

- Eventually, a remainder of zero occurs in the sequence of terms: $a = r_0 > r_1 > r_2 > \cdots \geq 0$. The sequence can't contain more than a terms.
- By Lemma 1
 $\gcd(a, b) = \gcd(r_0, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n$.
- Hence the greatest common divisor is the last nonzero remainder in the sequence of divisions.



Étienne Bézout
(1730-1783)



gcds as Linear Combinations

Bézout's Theorem 贝祖定理 : If a and b are positive integers, then there exist integers s and t such that $\gcd(a,b) = sa + tb$.

(proof in exercises of Section 5.2)

Definition: If a and b are positive integers, then integers s and t such that $\gcd(a,b) = sa + tb$ are called *Bézout coefficients* of a and b . The equation $\gcd(a,b) = sa + tb$ is called *Bézout's identity*.

- By Bézout's Theorem, the gcd of integers a and b can be expressed in the form $sa + tb$ where s and t are integers. This is a *linear combination* with integer coefficients of a and b .
 - $\gcd(6,14) = (-2) \cdot 6 + 1 \cdot 14$

Finding gcds as Linear Combinations

Example: Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

Solution: First use the Euclidean algorithm to show $\gcd(252, 198) = 18$

i. $252 = 1 \cdot 198 + 54$

ii. $198 = 3 \cdot 54 + 36$

iii. $54 = 1 \cdot 36 + 18$

iv. $36 = 2 \cdot 18$

- Now working backwards, from **iii** and **i** above
 - $18 = 54 - 1 \cdot 36$
 - $36 = 198 - 3 \cdot 54$
- Substituting the 2nd equation into the 1st yields:
 - $18 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198$
- Substituting $54 = 252 - 1 \cdot 198$ (from **i**)) yields:
 - $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$
- This method illustrated above is a two pass method. It first uses the Euclidean algorithm to find the gcd and then works backwards to express the gcd as a linear combination of the original two integers. A one pass method, called the *extended Euclidean algorithm*, is developed in the exercises.

Consequences of Bézout's Theorem

Lemma 2: If a , b , and c are positive integers such that $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Proof: Assume $\gcd(a, b) = 1$ and $a \mid bc$

- Since $\gcd(a, b) = 1$, by Bézout's Theorem there are integers s and t such that $sa + tb = 1$.
- Multiplying both sides of the equation by c , yields $sac + tbc = c$.
- From Theorem 1 of Section 4.1:
 $a \mid tbc$ (part ii) and a divides $sac + tbc$ since $a \mid sac$ and $a \mid tbc$ (part i)
- We conclude $a \mid c$, since $sac + tbc = c$.

Lemma 3: If p is prime and $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i .

(proof uses mathematical induction; see Exercise 64 of Section 5.1)

- Lemma 3 is crucial in the proof of the uniqueness of prime factorizations.

Uniqueness of Prime Factorization

- We will prove that a prime factorization of a positive integer where the primes are in nondecreasing order is unique. (This part of the fundamental theorem of arithmetic. The other part, which asserts that every positive integer has a prime factorization into primes, will be proved in Section 5.2.)

Proof: (*by contradiction*) Suppose that the positive integer n can be written as a product of primes in two distinct ways:

$$n = p_1 p_2 \cdots p_s \text{ and } n = q_1 q_2 \cdots q_t.$$

- Remove all common primes from the factorizations to get

$$p_{i_1} p_{i_2} \cdots p_{i_u} = q_{j_1} q_{j_2} \cdots q_{j_v}.$$

- By Lemma 3, it follows that p_{i_1} divides q_{j_k} , for some k , contradicting the assumption that
- p_{i_1} and q_{j_k} are distinct primes.

- Hence, there can be at most one factorization of n into primes in nondecreasing order.

Dividing Congruences by an Integer

- Dividing both sides of a valid congruence by an integer does not always produce a valid congruence (see Section 4.1).
- But dividing by an integer relatively prime to the modulus does produce a valid congruence:

Theorem 7: Let m be a positive integer and let a , b , and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof: Since $ac \equiv bc \pmod{m}$, $m \mid ac - bc = c(a - b)$ by Lemma 2 and the fact that $\gcd(c, m) = 1$, it follows that $m \mid a - b$. Hence, $a \equiv b \pmod{m}$.





Exercise

- Sec 4.3 13, 23